

ActInf Livestream #019.2 ~ Deeply Felt Affect: The Emergence of Valence i

Livestream | Deeply Felt Affect: The Emergence of Valence in Deep Active Inference | 1:58:09

hello

and welcome to the active inference lab

this is active inference live stream

number 19.2 on april 13th

2021 so welcome everyone and thank you

for joining

we are the active inference lab we are a

participatory online lab that is

communicating

learning and practicing active inference

you can find us at the links that are

listed here

this is recorded in an archived live

stream so please provide us with

feedback so that we can

be improving on our work all backgrounds

and perspectives are welcome here

and we'll be following good video

etiquette for live streams

today we're in the follow-up discussion

19.2

um our third discussion following on the

dot zero

contextualizing video and the dot one

first discussion

last week with casper and everyone else

and so today in 19.2

we're going to be continuing this

discussion on deeply felt

affect hearing some slides from casper

bringing up some of these questions that

we're having about implications

and other topics so really the goal is

just to

learn and discuss and we'll be going

through whatever we need to
and next week and beyond we'll be
discussing different papers
so let's just do a quick round of
introduction
and then we can see and
hear what casper has brought to to
share with us today so i'll start i'm
daniel

i'm a postdoc in california and i'll
pass it to
scott

hi there folks i'm scott david i'm the
director of the information risk
and synthetic intelligence research
initiative at the university of
washington applied physics lab
and i'll transfer to alex

thanks i'm alex i'm researcher in
systems

management school in moscow russia
and i pass it to blue

good morning i am blue knight i'm an
independent research consultant
based out of new mexico and i will pass
it to

dean hi i'm dean

i'm up here in calgary canada retired
trying to keep up with all this active
important stuff

and i'll pass it to stephen hello i'm
stephen i'm in toronto

i'm just working on a practice-based phd
i do a lot a lot of work with
participatory theater

and social topographies for exploring
community development

and i'll pass it over to i think it's

now casper is that right
yes so my name is kasper hasp
i'm a phd student at the university of
amsterdam
focusing on the emergence of
social cognition in
development and evolution um
and today i'll be will be continuing uh
daniel said from last time so i'll be
mostly assuming that you're already
listening to that live stream as well
but
just for for everybody's sanity
i'll give a short intro on what we've
been discussing
so far and just
to describe the context of this
research um
i was i was when i started this
particular
topic deeply felt effect for this paper
i was in the process of setting up um
the topic of my phd research
and focusing on social cognition but
then in doing that
um i realized that we were seriously
lacking
um there were serious gaps in the
literature on how we actually connect
from our bodily existence um to
like abstract social world and our
expectations about our social world
shape the way we feel physically and how
that
recursively shapes the way we think
about our social environment
um and that's where i actually got into
this

kind of well you could call it a rabbit
hole
of affects and emotions um
because i think that affect and
is basically the kind of through
emotions is what links
um the social world the abstract social
relations that we
are you know experts at innately you
could say even to a large extent
if you look at the efficiency with which
children in elementary school
kind of track each other's popularity
and whether they're part of the group or
not in this kind of
um i mean of course it's um
we're not making i know like this
dichotomy of nature and nurture it's an
emergent system
it's like there's an interaction going
on um
it's you can say that we're you know
one of the things we teach each other to
do um
and that's where our system is kind of
prepared to do
in a large uh to large degree is to
track
events happening in our social
environment and
from an evolutionary perspective that's
a very logical thing
to do because for hundreds of thousands
of years
you could say our each of our yeah
our survival would be
entirely dependent on our capacity to be
part of the group

so to speak um our capacity to function
in a group context
not just on the individual level but
also on the group level so
our ability to survive as
a species dependent on our capacity to
coordinate and distribute tasks
and specialize in different directions
so uh in the very same way
shaping and basically forming a body at
a higher level of organization
and just like the cells in our body are
differentiated in different directions
we as humans have to differentiate
ourselves
to form one large functioning body
and you think like what does this have
to do with the paper well
um essentially
the idea is that um the affective
states as we talked about them last time
in terms of kind of subjective fitness
we focus largely on the affective states
of
a single agent doing some tasks in like
a
teammates but today we'll kind of move
towards
a steps beyond that and in particular
the way in which tracking
the effective states of others in in the
body let's speak
in the unit of organization as a group
and how that's tracking can allow
you to to find your place in the body
just in a completely analogous to the
way
in which cells track each other's stress

signals

in the body so the pain

um basically the empath our capacity to

empathize with others would be very

similar to the way in which

cells respond to each other's

stress signals uh yeah stephen

i'll just be curious to know in your

sort of journey

often affect is tied to sensory feelings

you know at the kind of

personal level and so so that it becomes

a kind of a

a macro social sort of measure

was that something that you already had

in mind or is that something that sort

of

evolved as you were exploring this

these questions yeah i was essentially

um interested in that connection between

the very strong uh like experiential

aspect

of effect like when you feel embarrassed

you're literally

getting uh you can get a

red face basically so it's like a

very much an embodied aspect of that

[Music]

and somehow you're able to connect that

to let's say what other people

how um yeah your beliefs about how other

people are perceiving you

right so um that connection between

the bodily states and those abstract um

beliefs about the social environment

is something that yeah has kind of

driven this

and this effort and scotland yeah

one thing that's fascinating to me is
that
the notion that there might be multiple
levels of that harmonic coupling
right and multiple pathways so that the
dynamics of it
it's certainly you know it everything
every relationship and interaction is
describable
potentially as a binary and then
then you can elaborate on it right and
so there's
is all sorts of nested affordances
that have direct and indirect qualities
to informing
those interactions and so that
picking apart and disambiguating the
general relationship is fascinating
right so that and the and it feels like
just
anticipating the surprises that active
inference continuously brings us
that there's going to be a i guess i
would not just say a scale
independence but also a local
independence
to some aspects of it so what i mean by
that is they'll be
analytical interoperability i would
expect
more broadly than we might get with
other analytical framings
simply by nature of the manner in which
active inference approaches the
questions
of relationship so if you know what i
guess i'm saying in that is
i think it may be possible this is my

years as a lawyer
it may be possible to insert rhetorical
artificial or let me use the word
synthetic instead
um linkages in there and that's
something i'd be very interested in
understanding what your thoughts are and
how we can you know we
assert them into being right and we and
so describing them if it's efficacious
to describe them in a certain
way then they become adopted
those things don't exist in nature until
we describe them
so we have an opportunity now to
describe and disambiguate these things
in a way
that i'll use the it's like tempering
with bach
and the well-tempered clavier we get to
decide on the notes
or like when they decided on 360 degrees
in a circle because it divides well
i think that was the reason the
phoenicians or whoever did that
um so here we get to just maybe get to
decide without doing an injustice to
reality
on how to parse the linkages anyway
it's it's exciting because i love the
way you you introduce that possibility
of starting to pick apart those things
and call and name them
in a way that starts to do like david
boehm stuff where every reality is first
a thought
and here we're having the thoughts so
thank you for inviting us into that

space it's fantastic
thank you thanks for joining yeah um
we'll get more into that in this second
but i agree like it's kind of
refers back to what i talked about last
time in terms of
compositionality or the kind of lego box
in a sense that
um yeah these abstract relationships
can be applied on different levels of
organization
and that gives you a freedom
[Music]
in a sense that if you're able
to capture this relationships on and the
way they talk to each
other on the different levels of
description that's
yeah that's really essential i think
also blue was
putting up her hands so so
i'm anxious to see how this model
unfolds
in tracking others i think that that's a
a great way to
add an extra layer but i wanted to kind
of ask first
about uh the model as we discussed it
last week
it's just something i've been thinking
about we
talked about how um surprise is
is related to like negative uh affects
right
so like you know whenever your
prediction doesn't track that that's
the the negative affects or or that's
the as opposed to

the positive uh valence and so i just
was wondering like do we ever
do it leave room for pleasant surprises
like you know if i don't go food
shopping
i don't expect there to be a chocolate
cake in my fridge but i come home and i
find a chocolate cake i'm like this is
awesome
so is there like a room for that or or
is it like
incongruent or how do you resolve that
yeah
in fact it's a necessity that comes out
of the fact
that we talked about last time that the
effective charge
only emerges when there's a difference
between the prior and the posterior
so like uh you can be pleasant you can
be positively surprised
your predictions can be better than
expected uh essentially so
or the let's say reality can be closer
to your preferences than you thought it
was
um or like further away
let's say um so pleasant surprise is
when you get
a nice present from from your loved ones
or something
but then basically there's like a kind
of um
surprise in the sense that there's a
certain preference being met
that you hadn't anticipated um yeah it's
like
thanks thanks for the gift card oh wow

it's so much it's like another surprise
after you've renormalized it to the fact
that you were receiving a gift
you made an expectation oh maybe you'll
be 20
and then it's more
i mean there's that aspect and then
there's um
also purely on the information
processing level so to speak
when you're listening to music um or
some kind of
um something that's aesthetically
pleasing
there's often these kind of
stimulation have some patterns that have
inherent predictability
so like they're you could say there's
can be surprisingly easy to resolve
um so it usually starts with setting up
some tension
and then suddenly resolving that you can
see that in many art forms
in storytelling and jokes telling jokes
you basically have to
create the tension and then suddenly the
punchline resolves that tension
and that's exactly the unsurprising or
sorry the surprising part
of that reduction in the free energy
essentially
listen but yeah that's a very good
question i mean i think that's a very
common
misunderstanding also um
that's
yeah an understandable misunderstanding
but that's because

once you get to these hierarchical models things are often kind of uh get much more nuanced and you can't just say like surprise is always bad and it is surprised with respect to a particular generative model that specifies your ideal world so to speak that's that's what creates this negative effect but then once you on the meta level start to create expectations about how close you are to this ideal then once you get surprised on those expectations it can be negative or positive essentially anyway so that's a long winded answer to your question let's have um a question from dave and then maybe casper will go through anything you want to present because we could definitely just keep on peppering with questions i totally feel it so dave and then continue on you're muted dave continue apropos of blues uh question some of the older researchers on affect notably sylvan tompkins and yak pong sepp go into a lot of detail on the differentiation of the emotions and sensations and affects and drives and so on one of the things that tompkins did a lot on was the speed of change very specifically uh a a rapid change in a sensation may have

a

very different uh effect from a slow
change also that's reflected in the face
in the in what he considered inborn
facial expressions underneath all the
layers of acculturation

a rapid smile is some drum is
sometimes directly contradictory to a
slow

smile one a fast smile means oh you
idiot a slow smile means
oh you got it too

a very good question i mean there's uh
indeed

the temporal dynamics they come into
play

are really um really important as well
um and especially

because these nested levels they tend to
the higher levels tend to be slower
so for feedback to trickle down

let's say you need some time um so then
if it's a slower kind of moving change
then your top down let's say priors can
already adapt a little bit

before the next step happens

um i mean that's what you see

in for example in music where
to really make something give to
have an impact on the listener you need
to have some kind of

temporal slowness to the piece where
there's like an intro

and like different stages to really kind
of prepare your expectations

so to speak um or

actually the joke example is also a good
one like some people are bad at telling

jokes they tell the punchline too early
whereas like
don't give the tension time to kind of
emerge basically um yes sarah actually
just made a comment in the chat
um about how this model might be
associated with humor
and then can we think about humor in
terms of surprise only or
how do we think about these complex
layers of emotion like catharsis
surprise at different layers of the
model like you had mentioned
yeah i think humor of course
many different levels um there's this
really simple
jokes right it's like one question and
then and surprising answer
i i sometimes in lectures i use this
example like
how do you call somebody without a body
and without a nose
nobody knows
it's like a really silly one but it's
kind of nice as an illustration because
you can think
basically the free energy is uh
accuracy um sorry it's a
complexity minus accuracy right so and
when you get a question like that and
you get two concepts
without the body and god knows and you
try to find one
unifying concept and doesn't work
and then you get the answer and the
answer is both true false is accurate
and it's like a reduction in complexity
so you get like a double

bonus in this case um
but that's an easy one because the more
interesting types of humor i guess
emerge when you have like
complex storylines and there are certain
i don't know like satire and all these
kinds of things so that's that's where
it's not as easy to describe but i think
it's also possible if you had some kind
of
generative model of the social
environment and then certain things
are kind of inappropriate and then
certain things are so inappropriate that
they become funny or something like that
like uh yeah there's every country has
those comedians i think that they're
able to
to walk that thin line of uh
extremity and so there was another
question i think
um can we do one more question oh yeah
yeah yeah scott scott you want to go for
it
the question so much is just on on that
on the
measurement of the subtleties
of the free energy there
are books on rhetoric and i'm trying to
find the title here
but it specifically lists dozens
hundreds of techniques of rhetoric
like the juxtaposition of words in a
sentence does one come before the other
as priming
or whatever a conceptual linking of
multiple sentences
paragraph structuring use of foreign

languages

french tends to indicate certain thing

of style german tends to indicate

accuracy whatever

and so those notions of metrics

from there are things that can be

observed so they can be measured

so the metrics from the humanities let's

call it linguistic

psychology they are positioned

and ready to dive in to inform

those variables in that space right and

so that

is the next interoperability is not

going to be technical interoperability

we got plenty of that

it's going to keep going but what we

have is this deluge of interactions as a

result the

true exponential increase in interaction

volumes because of

fifth order effective moore's law and

what we need

is meaning now to be interoperable

and that's where we get to the drive

here of the

shared framings of rhetoric

right and so one other point a hundred

thousand years ago

we were one village in the horn of east

africa

we radiated out around the world we came

up with different food

different language different politics

different gods

etc and now the internet brought us

together for show-and-tell

everyone's got everyone's work and

active inference on each other whether
we know it or not
and if we name it as such it gives us a
sandbox to play in
at massive scales and i so i love the
way you're
naturally and i think appropriately and
not by analogy but by direct application
the concepts
bringing in our behaviors that are
beyond what are seen as base of maslow's
hierarchy and needs basically it's
beautiful you're doing beautiful work so
thank you for
inviting us into the space it's great
thanks
actually that comes back in a sec
kind of needs hierarchy of needs or
yeah how to deal with that so that's
i i i just see one more question
we can uh until we find a break time we
can uh yeah
are you yeah well mine isn't really a
question
in this i'm curious whether or not
i have a question later but what i
wanted to find out was
when you discuss implications whether or
not my question gets answered before i
have to ask it
because i think from an effective
standpoint that's the part that's going
to get
really interesting so i just wanted to
put that out there
because i actually do have it written
down because it's a long one but i want
to ask it yet i want to see whether or

not

in the implications that you're going to
share with us whether or not
my expectations align with what you
think

we want to kind of find out more about
oh see then yeah and i'll try my best to
i mean it's impossible to be exhausted
but uh

i will try our best yeah um actually
about the book the rhetoric of style
yeah

i there's actually interesting another
example

this is like um the art of uh movie
directing

like choosing the shots and which shots
are like

kind of cognitively fluent so to speak
and if you're creating a horror movie
you like the exact opposite of that
and the certain ways you can create
anxiety

or like different kinds of emotions
based on how you

how you move the camera around and which
parts you do and don't show
those kind of

and one one reason to just this book in
particular just

advise that anyone take a look at this
if they're looking at this it has
hundreds of different eye examples each
one's one page it's like

sentence construction sentence basics
etymological things

tropes blah blah blah it just goes on
it's like a checklist but it's this

thick

and so this is the things to measure for
influencing

other groups and so this is i really

this is gene faunstock

f-a-h-n-e-s-t-o-c-k i really thought

this one was a really helpful one from a
checklist and systems approach because

we're really doing processes here

trying to anticipate those processes

that we can make into alcohol so

good stuff um

uh what was the title rhetoric or style

someone's describing that

rhetorical style the uses of language in
persuasion

by jeanne faunstock f-a-h-n-e-s-t-o-c-k

and in fact one of the things just we're
doing in our program

is we're trying to come up we're not

trying to come up with coming up with

metrics for business operating legal

technical and social variables

so we take technically feasible systems

like what we're talking about

generally here ai and others and then

test them for bolts reasonability

and the way we get there is just by

rhetoric

because we it's storytelling it's folk

ways you know they're just folk ways for
the future

and that's why the active inference

gives us that framing it basically says

you tell us what your things are in your
minds

you group and we'll take them and

synthesize them and feed them back to

you and then you get to decide whether
you like them or not
and so it's just really an interactive
element so really

thanks ella you're really interesting uh
points yeah

um all right so let's and let me try and
get to the um
the actual slides that i was gonna share
but i'm sure we can
find many of these uh slides

[Music]

um let's see how can i share my screen
on the bottom left the middle button
yeah yeah and then i'll resize it so
it can look however it looks on your
everybody's

computer i'll make it look normal so no
worries about that

okay um looks good

yeah okay so just to kind of recap a
little bit but i mean we already
already did that but last time we're
talking about this

nested set of um inferences essentially
that

happens or appears to happen actually
i'm pretty much uh

to kind of hint at the instrumentalist
discussion i i

i'm pretty much fine with completely
being completely

and so that i don't i don't mind if it's
um

as i said yeah i i like to always say
that all models are wrong but some are
useful

so i think this is a very useful way

looking at it so it's instrumental

whether it's true in a realist sense i

i don't really care that much i guess in

the end but

[Music]

mostly because i just don't have i

i don't think there's a way to make that

[Music]

claim um

and there's no way to prove it i mean

there's a way you can believe in it or

something but

um and you can say that you require some

amount of evidence to believe in it

but as long as it

works i i guess um for most

applications that i'm interested in or

maybe all

applications doesn't really matter

whether it's

yeah what's what the philosopher thinks

about whether it's real or not

as long as it works um

so that's my um i don't that's why i

always say i don't have a dog in that

fight um but

yeah just to recap um

a little bit too last time we talked a

lot about valence

but there's also other types or

dimensions of affect

and some of them are like uh there

you are like three fundamental

dimensions

like arousal being like kind of your

your activity um level your current

amount of um resources that your

your body is expanding and for

the current um task at hand or current
context
and power control so the perception of
whether
out of agency you can say whether you
are
in control of the situation or not
it's not exactly the same as valence in
a sense last time we talked a lot about
[Music]
control as well but you can have a
highly
positive experience let's say if you're
[Music]
able to if your trusts that you're
that your needs are being met even
though you don't have
control over them yourself um so there's
these different dimensions that to some
extent can
can correlate with each other
[Music]
and but don't always and you have
another one is something like approach
and avoidance
which is often related to valence but
you can think of cases like um anger
where you have an uh or aggression you
have an approach reaction
but you're angry um you're so you have
negative valence
high arousal um and an approach response
so there's these different dimensions
that sometimes correlate but not always
and there are ways to
let's say in very simplistic organisms
there's yeah the dimensionalities very
much reduced

and
when you're let's say a baby and you
start out with very simplistic
dimensions you feel content or you feel
discomfort and as you get older you
start to get
more of this this dimensionality starts
to um
become more nuanced so to speak and it's
hard to say and then
the way um
i'm relating this to to take this one
step
further um is to
think about what this means
for um when we have internal generative
models
what's that what those generated models
and what kind of influence they have
on your effective state your current
affective state
um and that's where it gets an
interesting application
in psychiatry because
as we all know there is a dark side
to our imagination and that is that
imagination induced induced effect
plays role in all kinds of mental
conditions and also just on everyday
life and
so these are the extreme extremities of
course the psychiatric
diagnostic condition categories but
everybody i think to some extent has
experienced
at least a few of these types of
effects inducing thoughts
and basically i would i can just run you

quickly through them but
imagination of imagined threats and
flashbacks to some
bad experiences in ptsd cravings so
that's
like imagining obtaining of
drugs that you have that you're attached
to
but can be also other types of addiction
[Music]

ocd is obsessive disorder is a really
really strong example people basically
kind of you could say they're the
effective responses that they
get when they have certain thoughts
are reinforcing the thoughts
so they are afraid of breaking their own
rules
and they they get this kind of dread of
breaking
them so there's like a cycle and this
kind of
cycle is very self-reinforcing
cycles of uh affective states
and that then guide the way your
imagination
works and then that triggers more
effective
responses in depression and reaction i
mean you can go
on i think almost i haven't really found
any
um disorder in psychiatric um
like the dsm that doesn't really
fall into this except for perhaps
they're really more like
um physical um problems but those are
often psychosomatic so

and there's also a connection there um
anyway so just
for as a very abstract analogy we can
construct a generative model to give
this some flavor and can be of some
social environment let's say you
start in uh you have four locations um
you can make this as complex as you want
but just to give you to give
for the sake of the simulation that i'm
gonna be showing
you can start neutral and so some safe
space
there is a place that's safe relatively
safe
but you get only a small reward and
there is a place that's a little bit
risky but you can get the high reward
and why is it risky well because it has
the probability
of transitioning to this absorbing
painful state
you can think of this as like um
asking somebody out on a date you have
the risk of getting rejected
or um as a hunter um hunting like a big
big prey can be dangerous can get hurt
and can also
do something safer but you get a smaller
reward so there's like
many types of situations in the world
where this
kind of discount structure comes back
where the higher higher rewards are
often associated with potential and
adverse events the problem with this
type of
state space is that it very quickly uh

explodes when you want to search into the future and the events as they unfold and it's something we call combinatorial explosion but let's say you want to plan ahead

and for a few time steps in this case four time steps you want to anticipate what's going to happen um very quickly it's gonna it's yeah the number of possibilities

just becomes

yeah hard to track and um

the types of generative models that i've been

working on and carl frisson recently uh with a paper with me and

thomas barr and danny jar hofner

um developed a new

yeah i was basically sophisticated

inference something i mentioned the last time as well

um a way to do this kind of research

without having the whole

thing become computationally uh

intractable

and um yeah i'm not really going into

the technical details here

but essentially the core ideas that use

instead of having the whole policy space

which is multiplicative so you have

in this case the policy space if for

each transition you have three options

here you have four so you get something

to the order of

four to the power four and

a little bit less in this case but this

is just for

for um uh four

sorry four to the power three um for
just for three time steps
and of course in reality we often
anticipate
um more
steps into the future just yeah
without even thinking about it
explicitly
so there needs to be a way to do this in
a way yeah that
basic that you prevent just overloading
the system
um with anticipating
too many possible outcomes
and what is interesting is that you can
design
a model in which the way you explore the
future
is guided by your effective states
[Music]
and this what i did for this simulation
where you have
this exploration of the future
is itself an action model in a sense
that you're
[Music]
experimenting internally with potential
features
and the way you uh trend yeah basically
travel through
the decision tree is guided by this
action model and because it's an action
model it's guided by your effective
state
and because it's the way i've linked
that in my research is by adding
having this action model precision um
that we talked about in terms of the

valence states

um so what you get

as an interesting example what i

simulated here or so

and model that can do overthinking can

basically is something that

you may have come across in the

um when you discuss cascade effective

inference a long time ago

because then essentially this model uh

action model precision is guiding the

way the three searches

is performed but then

at some point this agent starts to

imagine

a little bit yeah some negative events

and starts to um

basically the precision starts to drop

of that the way they explore in the

future

and they start start to become more

likely

to imagine negative outcomes

because when the precision on their

action was high

they basically had a confident kind of

course of action in mind and where they

would go

and would automatically down regulate

unlikely negative

events but once this precision started

dropping this

agent start basically imagining

these negative outcomes and that's made

the precision drop

even more and then this agent yeah you

can see

i kind of calculated in terms of the

fraction of imagined
events that were negative negative
outcomes
call it like threat perception or
something and
and it's just an illustration of how you
could simulate a process something like
what we could call rumination um
or at least like yeah a negative
consequences
produced by overthinking the situation
and that's interesting because it means
that you're actually
can also model too much and
you would think in machine learning that
the longer you run your
your model the better it gets but once
you get this kind of recursive
relationship
you can actually get a kind of breakdown
in

[Music]

in the way the future is imagined or
anticipated
you bring this one step further

[Music]

you can also actually i have a picture
here as well
um this one
so this is where um something that looks
a little bit like
muscles um hierarchy can come into play
um even especially when you start to
make this with an
extra layers but here i just

[Music]

connected this sophisticate affective
inference

[Music]

to higher level states that in this case

that describe

exhaustion hunger and valence state

um and the point is

and that's something i also highlighted

last time it didn't show the model

itself

that and the more hungry you are

the more strongly you're driven by

preferences for fruits

the more exhausted you are the more

strongly you are

yeah driven to prevent expending energy

and valence is then a kind of summary of

how well you're able

to manage all these different

kinds of preferences and here i just

show an

example simulation where you have these

three different

higher level states and they're kind of

yeah

valence er sorry exhausts the exhaustion

and hunger are kind of

competing in a sense because the agent

needs to

um to get foods they need to expend

energy

[Music]

but yeah while they're at home resting

their hunger increases the energy

levels yeah the exhaustion decreases so

there's this kind of

cycle cyclical behavior

where yeah if you follow it's kind of

interesting you see that

[Music]

there is hungry in this case a very strong driver of negative affect so you see that whenever anger is high telling states is low um and the best situations yeah are in this kind of cases where you see that both of them are low at the same time so they're basically

[Music]

not hungry and not exhausted either i mean there's different kinds of balances you can implement here and this is in that sense just an illustration but you can already think of ways in which this also relates to things like depression for example where these uh people with depression would uh often report or have that had experienced depression and would report that they um have a lack of interest a reduced appetite because if you're really exhausted that means that you have a strong preference against expanding energy and that means that you can't anticipate the pleasure of eating anymore and you can't even experience it because you have this extreme and exhaustion or in this case the express yeah the depression states would be associated with experience of lack of energy i mean this is like a an example of how different kinds of

phenomena that we know from psychiatric conditions actually emerge once you start to construct these types of models in detail let's see if i can so there's this aspect so this is kind of on the on an individual level and as i said i wanted to share with you how this works when you start to think about multiple um unless there are any questions um i'll just continue you can hit um hide on that stop sharing thing to move it and then i think let's have dean and then steven so dean go for it well my question fits perfectly with what you just talked about so essentially uh comparing a t-search to a tree search which i think of as natural order and your your slide about combinational explosion my question was in simple likelihood mapping the binary is what it is the explorer action is limited answering what is good and what is bad on some version of valence continuum past that one wonders about what happens when trade-offs become trade-ins this would add a layer of complexity by keeping what is good what is bad and what if to the mix in the teammates example there would be a third option trap door with optioning as function being the key that their door would

contain say
four times reward and two times pain so
not enough to cause permanent
consequence
but still exponential relative to the
acute outcome
there'd be no cue to this option only
the barrier that is either charged or
not is transparent
you can see the reward you can see the
lightning bolt
signal signaling that the barrier could
be charged
based on what if which extends the
epistemic and specific
of the simplified likelihood mapping to
now include distribution
which is third possibility and
perception sensing which is the
potential shock just as part of a 4x
result
do you think effective charge remains
continuous
in terms of the path to renewed
confidence
i know that's a really long thing but
essentially
i thought this was the direction you
might go with this work
and as it turned out way to go because
this is exactly the kind of question
when you have a learner
who's thrown into a novel context they
don't tend to only have a binary choice
so how do we model for it because
we want this to apply in real situations
so
clap clap that's perfect thanks

i mean um the continuous versus discrete
state space
problem is pretty universal yeah
um i think the way our minds
tend to deal with it is by course
grinding it's first in
like categories and then later on when
you're locally in the situation you end
up
sorting out the details on the
continuous level
exactly um so yeah that's i didn't
i kind of ignored the continuous state
space
story here and just for the sake of the
simulation
but what's nice is when you start to
look at the parameters actually you get
do get the continuous play or like the
sorry the way the variables
vary over the course of simulation you
get the continuous part back
so cool and also dean in that i i
i saw like you're approaching a t-turn
it's a discreet
choice in the moment but when you pull
back and when you include
what if it almost uh makes
continuous through the generative model
of the learner
in this case it makes that situation go
from discrete to continuous
and also when you're learning you could
rest or you could eat
and so it's actually like our affect as
learners
as we're optimally foraging it really
does need to be considered in an

embodied sense because
at any point you know when reading
casper's paper we all could have thought
wow
we're a little tired and then that
changes that's a binary decision but
it's part of multiple
overlapping continuums so yeah it's
reflect
reconcile recovery cool so thanks for
the question dean um stephen
go for the question and then um anyone
else and then if not casper
yeah i'd just like to get your thoughts
on how
context comes in i see we think about
scaling and it's very tempting to sort
of all this can scale in many ways but
this almost speaks to this idea of like
they talk about trans contextuality i we
need to think about different
ways that we're in the environment and i
think that the way that
also ties in with the psychiatric
challenges is
that like you say the affect
um starts to run away from itself so
that
unlike someone who maybe doesn't have
that disorder and can recalibrate or
re-attune
to the new environment someone with the
disorder
it's sort of like you say it runs away
from itself but i'd be interested
because one of the things with this work
is it it puts the environment
into the potential treatment regime and

it doesn't just make it all about the
mind and the body
so just context and and how that
that that has an impact
um yeah so
in these types of models in this case i
[Music]
simplified it but as we remember from
the last time
this d high level d would also include
contextual states and then those
contextual states can be inferred based
on other things like
external cues and
so for example me observing myself
um sitting at the dining table
and smelling foods could trigger
[Music]
a kind of hunger
state and then that increases my
appetite
basically or like increase my preference
for eating food so there's
there is actually yeah there is space to
include
um just things like
where the context determines
the way your preferences are structured
and
or at least influences the way they're
structured
so and this has then interesting
implications for things like
ptsd where people would get flashbacks
when they're
in a certain context that they
associated with this experience
for example or addiction as well it's

like um

people on the first line of treatment in

strong addictions

as just take people out of their

contexts so that these cravings aren't

triggered

[Music]

so and i think when you say trans

contextuality

that's a really interesting word i guess

what you mean

is that you have a different version of

you

for every different context in a sense

yeah and it's also it's it's that that

that that

piece i mean they're talking about that

a lot now with um

developing um gregory bateson's work

in terms of looking

at multiple contexts

and warm data they're talking about warm

data so data it's not just

dead data but is alive in a way the sort

of stuff we talk about has

affective content and trans

contextuality means if you want to

understand different contexts

and not just deal with them separately

and interrelate them then there needs to

be some way that they're

the living component of context is

somehow

held alive um yeah

yeah that comes actually maybe that's a

nice segue unless there are other

questions there's a nice segue to the

um culture part because

great yep go for it yeah okay

um because

the most stable part of our context

[Music]

besides uh basic physical variables like

temperature

etc but our body temperature for example

but

once you move beyond that and one of the
most stable

components is other people in our
environment um and

how we how we coordinate with them um
and it comes back to

also uh some of the points that uh
were made on the rhetoric uh story
but just on the i guess i kind of like
to start bottom up so i just start with
the way um the

yeah the affected states translate um so
shared affects

and and cultured emotions so in this
case um

i i construct a few gerund models
um i can have the ones that are
associated with competition
or with cooperation and there are
different kinds of models

and there would be um if i believe
somebody is an opponent
then i have a certain expectation about
what they prefer

if they're antagonistic they so in the
sense that they prefer

they like to observe that i don't
i don't get any reward for example
or they may be just competing in the
sense that they

just want to maximize their own gain and
different kinds of
generative models i can have for the way
other agents behave so that would be
like the

the antagonistic and competitive
versions

and then there's like corporation
um can also have different flavors you
can have agents who genuinely
have preferences for
maximizing your gain as well or you can
have

agents who only cooperate when it just
because it happens to maximize their
gain

[Music]

these kinds of preference vectors is
basically what you can infer
based on how people behave and what kind
of situations stay

on um something like an altruist
altruism versus egoism access
you could think of it like that but and
then beyond egoism there's something
like

um machiavellian no it's something like
you want the other person to have less
so that's something

i guess even beyond egoism and anyway
there's different kinds of uh generative
models that you can
apply to make sense of the behavior of
others

um and just want to share with you a
generative
like a dynamic model that i made quite a
long time ago before actually

i got into active inference
modeling per se and then it took a while
to get it out
published but this is work from 2000
2015. um
and it was a pretty big challenge to get
it published because people
yeah it's like dynamical modeling is not
very easy to
sell to just psychologists who are used
to their
t-tests and everything but this model
was about
child play or actually play can be any
interaction between agents actually it
was child play was more like
the immediate application but
what is happening here is that every
agent generates
a decision and just the binary decision
in this case um zero and one as
was mentioned before you can often
reduce it back to binaries um
and in this case it's about playing so
are they playing together yes or no
if the agent tries to play with the
other agent
then only if both of them are trying to
do that
they have the emergent play outcome is
playing together
so there's a kind of conditionality
embedded in their environment if one of
them tries to play together and the
other one doesn't then they
then you get a zero outcome and since
they both end up being low
this kind of um

builds in this dynamical model
context i built different um
self influences off on the cell this is
based on a
emotion theory by nico ryder i don't
know i mean
he's kind of famous in the netherlands
but
emotional theorists but then kind of
revamped
for dynamical system theory and the
basic idea is that you have these
concerns so
that was also referred to in this muscle
pyramid um and they these concerns
and tend to drive behavior
but they also tend to drive or tend to
generate appraisals in terms of whether
your concerns
are are being met and that can trigger
emotional expressions
which then influence the other
person in the environment and vice versa
of course
so that was something from previous work
um
but then yeah started to use this to
generate
um i mean
yeah translate it to activate
essentially um
it's pretty easy to do so you can do
this with almost any almost any
dynamical model so
if you're interested in active inference
but you already are familiar with some
other dynamical models
usually it's a pretty straightforward

translation

it just means that you're reinterpreting

certain parts of the mess

sometimes means that you're reorganizing

certain equations

but often the core notions underlying

model because i mean most yeah the

models are always

all i mean usually they're fought well

through quite carefully by scientists so

i

tend to think of every model as a

potential and

yeah that has something of value that

can be translated into the new

into this interpretive framework

and here it's about what kind of

preferences the agents have

and this is the lower level so i make it

fix

their preferences for being alone being

together and then

a kind of this is parameter here would

be something like a

symmetry bias so this is a preference

against

doing something else than the other

person

it's it's more of a hypothesis as i said

before every model is a

parameterization is a hypothesis it can

be tested whether there's such a

symmetry bias

in practice there does seem to be such a

thing

anyway just not to overwhelm you with

numbers but

we can then make this hierarchical and

let it play out over

[Music]

over time steps and then

this agent starts to be able to manage
their concerns

over time so some agents might like to
play together

a lot more than others so then you get
something like a baseline

um concern for autonomy or
playing alone versus a concern for
um togetherness or belonging or
something like that

and those would be i think in every
person are kind of competing

so the idea here is that you're
gathering evidence for each of these
states and they are

yeah they are driving your preferences
but when you start to feel too lonely
then you might be willing to sacrifice
autonomy and

when you start to um and vice versa
when you need and feel

you need more autonomy might be willing
to um

to play alone i mean this is a basic
concept it can be applied to any kind of
balance

being act between preferences

[Music]

and you can also apply it actually i
anticipate another slide here

anyway i i've applied the same concept
to model

things like trust so trust games

where you're trying to

get a sense are we in a context where

we're working together or are we in a
context where we are
against each other and it's actually the
same types of moles
but then with one additional twist is
that now you're also inferring
um yeah the
the preferences of the other agent in a
sense um
with the same models applied so there's
one i in addition to this there's one
final thing i wanted to
share let's see how much time oh we have
quite some time business
um i just wanted to
share with you a few slides from
to finish this from a talk i gave with
maxwell ramstadt
in canadian neuroscience spotlight was
near in september somewhere in september
um
just to go a little bit more in depth on
what you do when you
have this type of model that
characterizes some
psychiatric
[Music]
condition or could characterize it
so there's always a different kinds of
domains of the state space
and and we start with something we kind
of
presuppose a functioning uh
or at least the potential to function or
the model um
and that's the kind of starting point
but then you look at okay in what ways
can it break down

and what types of combinations of parameters result in different kinds of what we call computational phenotypes that could be associated with affective disorders

in this case and what's nice there is that

you can now start to do computational phenotype together kind of map of different

different kinds of problems that can emerge

and this map doesn't need to be strictly categorical in the sense that the parameter space can be largely continuous so you can get clustering but it doesn't have to be absolute you can code mobility you kind of get it for free because

there's like this kind of coupling and you can get yeah basically combine it with things like rdoc i don't know if you're familiar with that but it's

kind of um alternative to the dsm which is a category as mostly a categorical approach

our dog tries to focus more also in the underlying

mechanisms that generate these different phenotypes and in that sense

this type of computational phenotyping could be the underlying

statistical tool that could guide um different kinds of investigations of not just affective disorders in the basic

level but also social emotional

functioning

[Music]

because of course as we started out in
the beginning those are intricately
linked to each other and
that's where the context becomes
important

and the way this is the final slide that
i want to share
is that we can now use these types of
tools

let's say we can fit them to individuals
and based on what a wide variety
of data can be self reports can be
objective measures
be also things like
even the more traditional measures could
be used to kind of initialize the model
and there's different kind of different
approaches you can take there
and the basic point is that now you can
have

an um if you want to have a model
of the model right it's it's a it's a
metamodel it's a model of the model that
tends to describe

the way a person generates their
behaviors and their faults
and the way their feelings evolve over
time

once you have that you can use that to
employ

to basically forecast their weather
conditions so to speak

and those can be context dependent they
can be

too kind of tied back to a question that
was

posed on trans contextuality
we know that the chances of rain are
dependent on certain contextual factors
factors that don't really yeah that
depends on
high level dynamics
on the scale that are not local yeah
it's not local
[Music]
in the same way what's happening down
inside your
body or let's say your heart rate can be
[Music]
contextualized by what's happening in
the rest of the system
and in the end what's the kind of dream
for this
psychiatric application effectively of
an effective inference
would be that therapists and
psychiatrists etc
don't have like a static thing like a
book
like dsm that just puts people into
boxes but instead they have this dynamic
tool
that um updates itself
by through feedback with the clients
then
release that to help the therapists and
then
the therapists can also improve
that tool so in that sense you get a
triad um
where the and this can be applied not
just to
therapist client relationships but also
can be applied to

let's say help clients
within their home environment to relate
to their
um yeah their loved ones
their school and or their
yeah their work environment
this would be a
personalized treatment prediction tool
and treatment monitoring so there's a
different

[Music]

levels of application i mean i focus
here a lot on
psychiatric applications because i think
personally that's
one of them yeah one that i find
the most fascinating there's loads of
other applications

you can think of these types of

[Music]

models can be used for all kinds of
commercial purposes
if you think about entertainment
industry how to generate

[Music]

an output that is kind of personalized
um to entertain somebody
i mean there's so many so many steps
that you can take i
just uh focused on the psychiatric side
that i am
yeah that's pretty much it for now but
i'm further we can find some uh yep
nice rabbit holes too

[Music]

let me see how i can stop sharing yes
yeah awesome thank you casper super
great to have you just kind of help us

revisit the paper and also some of your
past work and some of this other work
and it really just struck me that you're
talking about moving beyond the dsm
into dynamic systems modeling the other
dsm

so it's a little acronym overlap but it
just shows how when we
have general principles it helps us
descend

into the nuance and the context the
subtlety of a clinical relationship
an entertainment relationship the
educational relationship

and actually do full justice to the
complexity of

the embodied situation rather than
thinking that we're going to just have
some sort of miso level theory
that ends up actually falling in between
sometimes

general dynamical systems theory or
complexity theory and then on the
actual specifics so it was awesome to
hear about these cases

so we have um stephen and then scott and
then anyone else who raises their hand
yeah thanks this is really exciting with
i've been i've been reading some of the
work around the um

the dynamical sort of options for
psychiatry that you've been doing and i
think

one thing i notice is you talk about the
self the other and an

outcome and like the outcome it seems
that

as people are getting immersed in the

outcome it gives a way to dissolve this
self other
dilemma which is in a way that the trap
of modern psychology and psychiatry is
you name someone the self because you
can analyze the self
and the other because you can name and
analyze the other and then everything
else just goes from there and the model
is just like
is it is dominant but never spoken about
whereas here the model center stage but
if you
start to make it about the outcome and
the context
now suddenly the self and the other can
start to take
a more dissolved space you know and that
and the
positive psychology and the client can
start to have more agency
in their own life so i just yeah i'd be
interested in just how you saw that
relationship to self other an outcome
yeah i mean
in the in the interaction the
distinction kind of dissolves to some
extent
in the sense that they are
co-creating each other's models so to
speak
um and they end up having like a shared
reality
to a certain degree um
and what makes this reality
real or when like shared expectations is
the fact that
they have them uh together um

and many of things in society are
actually function
only because other because we all to
some extent believe in them like
money is a very clear example like the
only value money has is that
other people believe it has value
and again i mean self other i guess you
can't really

escape it in the end um

i don't think we need to because
if it's just a matter of the level of
description i guess and just like the
cells

of the body are clear units but at the
same time

they are part of one hole and
yeah this is like the gestalt basically
this tall psychology kind of comes back
to haunt you in a sense
and there's the

emergent hole that's other than the sum
of the sparks

i don't really mind talking about the
units as long but i think what you
describe is really important that you
keep in mind that there is like
it's just one level yeah in the end a
reduction

i think there was another is it's a
follow-up question

[Music]

yeah well i think that makes sense i
think just like you were saying if
if the outcome is being measured
does that change the way that we think
about

like if we start looking at the space

the environment the outcome the context
i mean are we in a whole different
paradigm for treating
disorders that's a really interesting
one because
i mean i know some really interesting
research on the way in which
psychiatric categories have influenced
cognition and behavior across cultures
basically people believing they have a
depression
or they have depression and then they
can't get out of it because they
ended up believing that that's their
disease like a kind of discreet thing
that they just stuck in
so that's a really interesting example
where
in that sense are kind of like mind wear
and they program
and they kind of specify the way you
structure your world
and what i like about this kind of
individualized modeling is that you can
move away from those categories you
don't even have to
tell a person that they have a
clinical diagnosis you can tell
how you know how to help them because
they have individualized predictions
for treatment but you don't have to tell
them like oh
we're putting you in this bin
[Music]
there's somebody yeah that's like it
becomes like a spectrum and there's
different
attractors in that state space

but we don't need to
make these strict dichotomies anymore i
mean maybe for the insurance that's the
only
limit it's like the insurance companies
want to know do we have to pay for this
or not
it it's so true that the implicit story
of diagnosis
is you know you got two kinds of people
the diagnosed and the undiagnosed and if
you have three of these five symptoms
you're in this box
and it almost made me laugh at this last
point about insurance
how about uh pay per affective charge
let's
uh continuously tune that dial if they
just need like you know a little bit
pay a little bit or something there's a
whole level of nuance
that's when we're thinking about this
in a in a big way so thanks for that
answer um
scott go ahead casper then scott
it was something i also i was trying to
i gave a talk in my psychology
department or like the brain cognition
meeting in my
university whereas basically saying that
in principle if we can alternate this we
can treat everybody
and to the extent that they're
interested in like
improving aspects of their life i think
everybody has some things they would
like to improve so there's like
don't even need to just limit it

to like the kind of model where you just
wait until things go horribly wrong and
then you start to try
to help like preventive treatments
that are not based on the sky and this
kind of disease model essentially
anyway and those prodromes might be um
revealed through these inferred
parameters like uncertainty
so before the real situation begins it's
really

a good point so scott dave and blue
so fantastic there are so many things
so many threads here let me pull a
couple together and it's
on this

[Music]

space of social and internalized
norms and folkways so dsm
iv used to list the cultural phenomenon
that weren't considered
disorders like tarantella or you know
the

or the different things that you know
they were cultural not considered
disorders

but that always seemed to be very um
imperialistic

uh the way they did that but it's kind
of interesting when you start to move
into

the um you mentioned

insurance companies and what they will
pay for the naming

of disorders is is

many of them are just call disorders

because they don't let you

function as an industrial worker you

know what i mean there's all sorts of
context

outside that dictates whether a disorder
is a disorder

and so i love that notion you're talking
about about the clustering
and the predictive elements there's an
um

people in public health non-mental
health physical

disease health talk about forty percent

four zero forty percent of the

global health effort for any emerging

pandemic is repeated from one to another

so imagine the cost savings and resource
savings

if you start to do predictive mental
health like people have been talking
about the covet is causing the ptsd
generation so you can have these
clusters of

things that need to be dealt with on
scales of billions of people

and that idea about merging treatment
into norms so really not talking about
treatment you're just talking about
a continuity of existential support
in in society that's that really is
fascinating

here's the question so it's kind of an
interesting

there was an article on soft paternalism
a number of years ago and the economist
that idea that for instance if i have a
gambling problem i can register with the
state of missouri to not let me into the
gambling ships that go up and down the
mississippi

when i have a moment of clarity i can
say don't let me on the ships
and then i'm registered there and then
they won't let me on the ships when i'm
feeling the urge to go gamble so there's
this kind of
inviting that externality in one of the
things i wanted to
ask you about is your thoughts your gut
thoughts on sovereignty and so i
define a sovereign as something that
doesn't need to ask for permission or
forgiveness
and it's always an external teleology
it's always a projection out whether
it's a deity or royalty or a country or
a company
we project it out so we can self bind to
it okay that's my assertion
okay so i'm looking it's that
sovereignty has to have an abstraction
of its subjects
because that's the notion of so we
project out
it to be abstracting us so we can deal
with the big problems
and then the anomalies we take care with
of with a mass
customization of localizing it
right so it feels like your clinical
notion about psychological states could
be employed for political purposes and
i'm talking about in a positive way
not in a bad manipulation but a good
intentional manipulation
of affective states that are positive
for civic engagement
and then you basically use the same

notions as
psychological disorders but you start
correcting for let's call them political
disorders
right because it's just normative states
right
so that that's what i'm curious about is
have you done any thinking about
how to project cultural states from a
group
like like projecting it on the screen
and then everyone watches it and says
huh that's us up there let's go move
towards that now
sort of a question it's a really
interesting question
um i have a student that works on like
communicative
cultural dynamics assimilating the way
in which beliefs
kind of in a kind of minimalistic way
that can be seen as
um yeah modeled quite similarly to
viral spread of um but then
they're from the perspective of the
beliefs
that are spreading it's kind of like
every time it's trying to be
it's somebody tries to transmit it to
the next person
the question is how well does it fit
with
the next minds and only then
only if their fit is close enough
it can be it can reproduce essentially
so then your question is kind of like
one level beyond that where you're like
we're trying to predict which message

will fit

to which group of people and then that

can be taken in very equal

directions but you can also kind of use

it maybe to break

um uh political bubbles

and it leads to extremism something like

that so there's different kinds of

because in the end it's a tool um that's

right

and one note on that political side

because something people bring up is

we're not simply talking about the

communications or just the rhetoric

display

it's actually about doing a systems

identification and potentially finding

latent areas

of agreement that are just two terms

being used by two communities for

example

so yeah but very important and

interesting

topics i was just wondering uh sorry you

talked about sovereignty

so i was wondering how you view that

connecting to

sovereign because the way i see

sovereignty is just like

uh server and body is like something

that has

complete power over a certain domain and

then

talking about people within that domain

so they are completely

under the sovereignty of that governing

body

right the way the way the link happens

is my assertion is that
all sovereigns are projections
so they are they're they gods um
i don't say we invented god because i'm
an atheist but it might be a blasphemy
to other people so we invented deities
because there's different things you
know so it's like
we invented royalty we invented
countries we invented companies name
another sovereign they're the things
that don't ask for permission or
forgiveness
we invented them so that we can organize
at larger levels now we didn't say it
explicitly but that's
why we did it and so all of them are
projections
so they're all stories from us now they
take on a separate life
sort of but they can never be identified
that's the thing about sovereigns
there's always a metaphysical element
they can never be identical to the
subjects
that's why you that's why when a
sovereign dies in office it's a it's a
real
difficulty for the sovereignty
when a human who represents a sovereign
dies
because it has to be separated out you
have a state funeral but then the body
has to be separated
from the body politic right it's hobbs
it's a leviathan thing
leviathan can't be composed of the
individual parts it takes on a separate

nature
and anyway so that's why this is so
interesting that projection
and that act of collective projection
is totally where we are right now think
about all the problems
climate change global resources water
resources finance
things that can't be done by nation
states we need new sovereign levels
active inference is going to be the way
that we synthesize them
it will be whether we call it active
inference or not because it's all we've
ever done
as biological then living and social
organisms
and it's a way of framing it and so it's
a way of now to me
it's a new teleology we can project
forward just like you're saying
predicting on
psychology i think we can predict
politics and social states
because we can use predictive analytics
of clustering it's the same thing
you take a bunch of americans another
student who was modeling the elections
american elections uh with the same
um but i i like what you said about
it's kind of subjective it's like the
attribution of power
um but then it's like a totally
imaginary
thing just like we attribute power to
money
yeah yeah i mean money is is de-risking
consolidator

you know it is one form of money wixel
and smith and hume they all have
different ideas about money
but one way you can look at it is a
de-risking consolidator so it takes all
risk and makes it into one risk which is
how can i get more money
and if that's the case it's an
interaction de-risking tool
right actually this could give us a way
to discuss
um to bring it also to the earlier point
of instrumentalism
and realism because the fact that money
has instrumental value doesn't make it
real
i mean that's right that's like kind of
interesting question that i don't have
an answer to i mean
i guess in practice it's real because
people believe yeah it's like a queen
singing and one thing one thing i just
suggest you look at uh somebody just
surprised me a couple of months ago
about christopher alexander's work in
architecture if you haven't run into
that check it out
because what he does is basically looks
at patterns in architecture through deep
time and
and through space and the idea is that
existing practices are captured in
patterns of practice
and if you want to look at a solution
for an earlier for a problem
look at patterns of practice from other
situations and bring them
into the situation here similarly it

feels like we can in your work it's
revealing
that we can look at patterns of practice
and then take them as predictive models
and so that's what chris alexander does
that in architecture
and it's very straightforward there
because it's physical instantiations
like an archaeology
of the mind of architecture in a way or
the archaeology of the problem space
anyway something to look at it might be
uh fruitful thanks
thank you scott so we have um dave
blue and then a question from the chat
so go ahead dave
yeah um if you could uh
turn toward the realistic side of
your experience with with these
questions
do you kaspar have the feeling that
people's real models
are of the same structure
as these transpersonal
models or person to environment or
organism to environment
folks in this field seem to kind of uh
assume that oh yeah it is all turtles
all the way down or it's blankets all
the way down
and i don't know that i'm convinced that
which what's your intuition
interesting question um
i think that it's kind of similar to the
way
um
you can't last time we talked about
dimensionality and you can project from

3d

into 2d space but like understanding a
3d space while you're in a 2d
space it's kind of impossible
essentially in the end
and it's a little bit similar like
looking down
on the hierarchy i think it's much
easier makes things
feel much realer more real because
essentially they can
they're within your
yeah your mind can grasp them um
and when you move upwards you're sort of
the transpersonal
part i think it's um
our minds don't have yeah the capacity
to
fully grasp something like that
we can so it doesn't have the same
experiential
yeah i would just say

[Music]

detail maybe um
so yeah i think you're getting at an
interesting point where there seems to
be a difference between
the way what experientially
feels real and what
in it from an interme instrumentalist
perspective
and there doesn't seem to be much of a
difference if you can move through
the trans-personal level

[Music]

just like as i said i think
in principle humans can form
but higher levels of organization

can form bodies together and
every human is this unit
in this body just like every cell
is and every organ in the end um
has its own local experience so to speak
its own local data
maybe that is being integrated
to inform their states and their
decisions
but then there's the global pattern and
that's the local elements never have
direct access to it
and i think that's something that
gets at this point as you're describing
um but yeah i guess i don't
have the real answer just but i
understand experientially what you're
saying
in the sense that definitely there are
things that feel real
there are things that don't exist but
feel real that's also
that there are ways in which trans
personal
concepts can start to trigger
hallucinations in people for example and
there's all kinds of
[Music]
hedge cases and does it answer your
question i think it's a really
interesting one i don't have a good
answer
it does yeah i think it is an elephant
in the room
perhaps um yeah and if i could just run
off on a tangent
on that um the relation between inner
in the sense of what's inside a

fristonian blanket and inner in the sense of what's going on in various parts of the nervous system or the organism i tend to come at this by thinking that what the the physiological uh the anatomical correlates of what we've been focusing on for the last two weeks are largely going on in the midbrain the brain stem some of the lower structures in the cortex but the modeling goes on in the neocortex so the inside is actually executed outside in the portion of the nervous system that's really good at doing algorithms and pattern matching and reinvestment of pattern transformations but the purpose of doing the whole thing is largely especially in a small child or a non-human mammal outside the cortex so that probably blows the whole thing in principle yeah from the models that i built it usually doesn't really matter where the data comes from like uh can be inter internal signals external it's more important what the inference is about so like um extraoception is is perception about external states of the world an imperception about internal states of the system but i can have

extra external signals that may give me
information about my internal states and
vice versa

so this distinction is pretty important
but yeah that doesn't entirely resolve
the issue

just i like the fact that the models are
kind of agnostic about
where exactly the information comes from
cool a lot to say on this um blue

and then a question from the chat and
then anyone else in the chat or here who
wants to ask

so i loved the diagnosis or the
elimination of diagnosis
and clustering instead of of diagnosing
people except like when it comes to
insurance and so

casper i'm going to charge you with
obliterating this

in the minds of of insurance companies
also because i think that there's this
fundamental thing that happens

and i mean maybe i'm wrong but if i am
you know if i display some behavior of
you know someone with adhd and i display
a lot of those behaviors

so if i display some of this behavior
but then i go

see a psychiatrist and i'm diagnosed
with adhd

i think that there's a large degree
especially in younger
children like as they're coming into
themselves and dealing with identity
crises teenagers and so forth

i think that there's a large degree of
of excusing or

perpetuating my behavior because i have
a diagnosis

right so so if i'm diagnosed with
something like adhd like maybe that will
you know give me i'll give myself

permission or excuse

or or i'll cluster more strongly

into the group of people that have that
those kinds of behaviors

because i have a diagnosis i think that
this is very plausible

what do you think yeah that's also what
i was trying to

get it in a sense that people tend

to yeah use use these internal
representations to structure their

experience and then to

respond yeah to generate their responses

and then these boxes become
self-fulfilling prophecies essentially

um to a certain extent

then there's that side but there's also

then there's another side to it is that

there are also positive reports that

people when they can

attribute the effects to something

it makes them feel less anxious so they

can kind of explain

the way

[Music]

and yeah i don't know what's yeah it has
downsides

and upsides i guess yep

a very sensitive and interesting thing

but it reminds me of the 12

steps and the first step is to

acknowledge the kind of person that

one is obviously we're not going to go

into this and it's for others to
investigate
but it's a trade-off between having a
narrative certainty
a diagnostic confidence versus the
alternative
and so there's our teammates in meaning
or a narrative space
which one of these bifurcation paths
narratively
will we then have the lower levels of
our model like
whether we exercise or not or whether we
have a job or not those kinds of things
come into play
um i just want to ask the question from
the chat and then um
dean and stephen and scott so in the
chat someone wrote
can free energy principle based models
reach a level in the future to be able
to precisely predict
when emotion or valence happens and more
importantly when
in the temporal dynamics of that cycle
does thought subside
so how is our experience of thought
related to some of these simulations
with uh you know the countervailing
affect of chart or countervailing you
know
hidden states where does our experience
of thought come into play
i mean um it's to some extent
already being done in the sense that um
there's applications of in amsterdam
they have
uh they're trying they have been

treating ocd
with the brain stimulation and they
identify the locus
of the intervention using
fmri and fmri is driven by dcm
which is driven by free energy
minimization right so we're already
doing that to some degree
identifying the kind of
recurrent self-destructive
loops of behavior in in the brain
essentially
so internal behavior of the system
and then identifying ways to break break
those loops
so they they implant they just try to
find the kind of

[Music]

note at which the loop is um
through which the loop is circulating
and then they break yeah they
disturb that note once in a while and
that
actually leads to an improvement
in function it's kind of funny because
when they remove it
and loops tend to return pretty quickly
again so
seems not to be an it's it's kind of
it's not a pool treatment it's more like
a
prevention it gets more interesting if
you had like real time way
of getting into that then you don't
yeah you would need not just like um an
electrodes
in there but like a whole
a whole array of electrodes that are

like tracking
and it's a little bit scary but in
principle you could have like a
whole weather tracking
system inside and then um if there's a
hurricane emerging
and blocking the i mean they tried to do
that already for people with epilepsy
and
i guess then we all become like a cyborg
or something if you start to go that
and that's what maybe not
i mean what i guess i find more
interesting or like more
appealing is um
if we start to become more aware of how
these feelings
are
generated by our relation with the
external world so
there's this use case of um

[Music]

tracking um warehouse workers
that i've been uh seeking to apply some
of my work to

[Music]

where you could track their affective
states over the course of the day
and every person has a different
chronotype every person has different
times of day when they feel most
energetic and it depends also on how
well they've slept recently et cetera
and so if you had some intelligent
warehousing system
then you could base you could distribute
tasks
and the schedule of the warehouse

could be adjusted to suit
the preferences and to maximize
[Music]
well-being of the
the workers and potentially in the
process
also increase productivity so that's
like kind of capitalistic
potentially if we could construct things
like that
and we could use our modeling of
these effective dynamics over the course
of
day by day for each individual we could
use them to
care for people to make them to prevent
burnouts

[Music]
and potentially make that appealing for
people who are just driven by money by
profit unfortunately they're
always in the end just most
companies are just driven by profit so
to make that realistic and scalable you
need to
show that also is good for the for the
business
um anyway so that's like some more parts
i have
in reaction to that question um there
was a sub component what was the second
part could you read that again
um uh more importantly where in this
temporal dynamical cycle
does thought subside
that's a very nice question
i wish i knew let's keep let's keep
clicking on it

i guess there are certain parts of our
system that are able to
disable are kind of contributing to our
verbalization
and the way i feel thought at least the
kinds of introspective thoughts
that uh it's verbal um
tends to be a kind of
internal internalized social
skill essentially you first need to
learn
about how to speak with others and then
as a child i guess when you're really
young at some point you start to be able
to
have these dynamics run internally
and my suspicion is i think there are
some research actually confirming this
to some extent but
in order to create the verbal thoughts
you need to access different
pathways you just
attenuate the actual motor execution
um that's how they tend to think about
it in the literature
um and then there's imagery thought
which is yeah in a sense
even more amazing maybe because that's
not something you can
i mean verbal uh that's an actual action
that you do
so it's kind of easier to imagine how
you could have this action evolve
like unfold internally but
imagery is not something that you
actually
do in life it's just something you can
somehow

you control your own internal

[Music]

pathways that you normally you would use
for

for having a visual sensation

your mental actions can somehow control
your visual

processing pathways which i i have a
hard time

imagining how exactly that i mean

i can build a model of that i think but

yeah how it works in

like biologically i find it quite hard

to

[Music]

well that's that's the instrumentalist

tango we can build a few model

architectures we could think about some

differentiating experiments we could

imagine some informative

observations go from there continue the

discussion from there

so um in our last little bit let's go

with

dean scott stephen

thanks uh real quick um you talked about

personalized stuff and

the loop seems to come back on politics

because i remember a couple weeks ago

the first thing i think i said to you

after high was we can't

unlearn this once we start using these

diagnostic tools we can't unsee it

um so there's a lot of politics involved

and people have this ability

to to see things one thing i

want one reference i wanted to mention

and be especially

i think helpful to scott if he hasn't
already read it about this personalized
versus political
is an author by the name of paul khan
paul w khan
and his book is called origins of order
and what it does
is it tries to bring those two ends of
that continuum
into some sort of a not like there's no
balance between the two but there's kind
of a back and forth
and it's very looking at rules but not
just
on a legal level but on the on
on much bigger scales so it kind of fits
nicely with
what blue said about the diagnostics
what we've touched on a couple times
about the instrumental stuff
but i still think that if you try to
apply a lot of this stuff to
which is what i did before it would i
used very rudimentary
ways of expressing this but i gave young
learners these tools
and boy was it political the results of
that because
they could just see things and start
doing things and there was a whole lot
of people that went
hey whoa whoa whoa they're getting way
too much power from this
but anyway that's just an aside i just
wanted to share the reference because i
think it really ties in nicely with
some of the stuff that scott was talking
about earlier and obviously

i don't think the political part of it
goes away no matter how much we want to
be
agnostic thanks
i mean yeah in a political context in a
sense
what we're trying to determine is what
what is the generative model that
governs the body
so like this conflict
and tension and tension resolution
and it's kind of like the price we
have paid for being able to
to organize it's it's the body politic
the leviathan as brought up earlier will
we turn left will we turn right will we
look forward will we look backwards
there's a lot to this multi-level
political approach
so scott and then stephen yeah thanks
so the it it is a big um it's like
moving a slime mold
but you know from outside the um a
couple of things that jumped in in the
last bit um
and and where i'm going with this is the
notion of a capitalist enclosure of
consciousness
with the categories of abstraction so
the painter kandinsky
and i've said this before it just
praised on my mind
painter kandinsky said that violent
societies yield abstract
art and one of the things i've always
wondered about is
is the reverse also true is abstraction
a form of violence

and so here one of the things that i was thinking about in the last couple of comments was symbolic synesthesia right we don't really have a perceptual synesthesia but we take symbols and we substitute them in for things so like if i drank tequila when i had a bad experience with something else a bad tequila experience with something else then i'll remember the bad tequila experience as associated with that other thing not that i ever had a bad tequila experience i'm just saying if it never happened so that abstraction you know we had it kind of becomes our prior whatever and it goes back to that notion of the the humor element too the abstractions are taking us to a place and then revealing the reveal comes right and so one of the things i was wondering about on the capitalist side we don't want surprise we don't want punch lines for jokes because that's risk right now now you don't have innovation then that's the problem is there's this tension between innovation and stasis right and so they want innovation but they don't want somebody else to innovate and take the risk but then they want the benefit but anyway so from a capitalist enclosure perspective one of the things i'm wondering is does

active inference and this is the
question
does active inference we we're all
taking care of each other's active
inference exhaust
right we're all taking the where
everyone's doing active inference on
each other and we're all the
externalities for everyone else and so
we're all having to internalize everyone
else's
discovery pathway in a sense right we do
we're loading on each other so that
space
of exhaust management is a commons in a
way we're co-managing
risk so when i talk about money
as a universal risk consolidator
might the metrics of value in markets
give us insights not entire insights
because markets don't tell you
everything not everything is measurable
that way
but give its insights into preferences
in a way that starts to give us the
insight into that
co-management of the active inference
exhaust
essentially so it's really treating
commons not as a fisheries commons like
with an asset commons but as a risk
commons
and it's basically consciousness as a
commons
so the imagine that the mind exists in
language rather than the brain
the brains are tuned into the shared
mind

and now we got to manage that share mind
so we get nonsense in there something
that's wrong
science or bad politics or not
functional economics or whatever
and we want to get rid of that because
it's spoiling the commons
so just that was what came to me that
this idea the sovereign
feels like how we synthesize an
externality to say whoa whoa whoa
we got to get this right because we got
some big scale stuff we got to do like
ants and bees can do without memos we
gotta we have memos so we got to write
the memos
so what do the memos say and that's why
you get to dsm and stuff like that and
what they're saying is just like
foucault said
pastoral control is what drives people
together with certain behavior
right and so it's that same thing are we
being driven
you know is it is are we being driven to
be industrial
workers or now we're being driven to
something else and how do we manage that
that commons of our infants active
inference
commons to aspire to what we want to be
now there's a question sorry
i mean it kind of makes me think about
the way in which um
the major religions have mostly emerged
as soon as
um at least during the same amount of
time that

they ended up being able to write it
down basically so it's like a pro
it was a recursive relation of course
but you
basically need a culture um

[Music]

for that level of unity there's two
directions you can go and i guess it's
probably both ways but on the one hand
you need to be organized enough to
have a culture that you're able to write
things down

and once you're able to do so you can
basically you can kind of hard code
the structures and you end up
having a competitive advantage over
other

potential models essentially so maybe
the mind

and culture are emerging phenomenon of
complexity

right maybe complexity causes the mind
because

we need we need to when we say we need
to do something

we often think it's in response to
something but maybe that thing itself is
another

instantiation of complexity itself and
so maybe what we're doing

is merely red queening the whole thing

right just keeping up we think we're
actually innovating and doing things but
really what's happening is

with increasing exponential increase in
interaction volume and this and this is
not as

this is what i actually believe in

pursuing

if there's a raw increase since the big

bang

in exponential increase in interaction

volume

maybe all we do is just keep up

and keep um dealing with that

externality

as living organisms in order to maintain

neg entropy states

maybe we just have to keep rushing into

higher levels of order

organization in order just to not

dissipate

anyway thank you scott

there's also opposite ways thinking

about like um

where these organized structures are

like the best

way to dissipate energy so there's like

a kind of

just like the convective bubbles in uh

in some kind of uh

heated um volume of the water or fluid

and the substructure basically helps

the system to dissipate the energy um

so there are certain it's like

interesting two ways of looking

it's the same thing in a sense it's that

structural

dissipation of risk is what you have in

law and that's that's all it is

it's just it's a time machine that

predicts the future

everyone can look at the prediction in a

contract or law it's enforceable and

they can rely on it

and so it saves resources so it's

exactly what you're saying

thank you um stephen if you want to make
a really quick last comment because
we're out of time and then we'll close
out

yeah thanks um i was just going to say i
mean sort of ties in this idea of
abstraction

and the um is i do a lot of work with
action methods
and ways of working in space and i think
it ties into this question i think it
might also tie into next week's
talk is there's certain things that we
can realistically

get a handle on if we can take a
perspective on it

so for instance you talked about
exteriorception

that sometimes it gives you information
you take a perspective on to reflect on
your internal states

and maybe you look at your internal
states but i think there is this danger
that this abstraction starts to take
over

because there's a whole load of levels
particularly you know when it's the
embodi which we can't see and that's
okay

you know these these blankets are
blankets can go down

and um it's beyond our
perception to take a a thought on
and the reason that i think that can be
interesting is um
is because in in action methods and
psychodrama

and these types of action approaches
rather than the freudian approaches
which basically
took over is okay you can't take all
these percep
perspectives and analyze it in the way
that we've got in this habit of doing
but these other ways of knowing
embodying
exploring i think that that
also is another direction which often
has not thought about so i just think
that um
as we go into this what you're saying
and the way that we think about
instrumentalization
is that you know we might need to
instrumentalize
using the the sort of the the more
embodied
effective methods and processes
as well as this kind of other approaches
because i think that
we will hit the limits of an analysis
and i think that might be something that
this conversation sort of
talked about so anyway i just thought
i'd put that there if you've got any
thoughts on that and i'll
thanks a lot for a great presentation
yeah i mean i think you make a very good
point and that's
uh and thanks for the compliment and i
mean there's the
the connection or the integration across
levels
let's say that's important for your
well-being then what you describe is a

very
logical pathway to improve
people's mental state in the end as well
when they're more connected with their
embodied
existence instead of living in that
abstract space
um i mean i think i have
as a scientist i have a probably
increased risk of going down that path
because you're always thinking
about things so yeah
um i totally agree that that's probably
many
treatments actually i mean that's why
meditation is such a
appears to be socially effective as well
anyway thanks for the thoughts yeah very
good point yeah thanks so much everyone
for participating and casper it's
you're not always thinking sometimes
you're coming on and hanging out with us
i guess but
maybe thinking during that too but
really um
fun discussion we hope to have all of
you on
again whether it's a paper you've
authored or not so thanks a lot
and uh yeah until next time